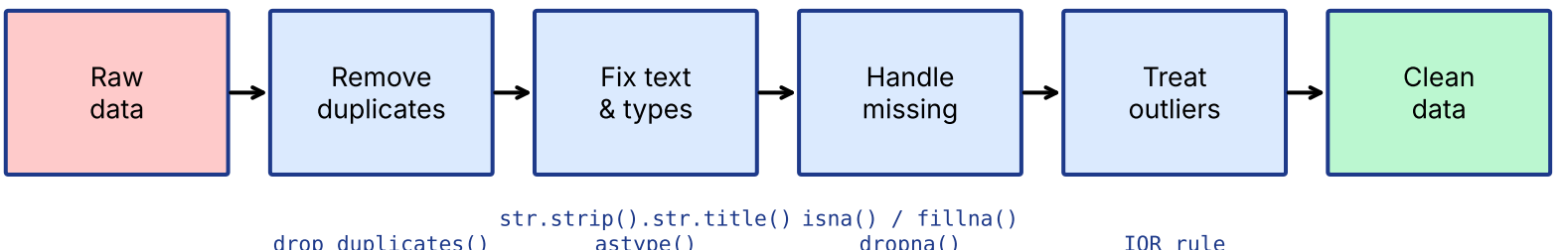The data cleaning pipeline